Logarithms and Their Properties

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Evaluating logarithms**

Evaluate each logarithm exactly:

a $\log_2 32 = 5$ **b** $\log_3 \frac{1}{9} = -2$ **c** $\log 1000 = 3$ **d** $\ln e^7 = 7$

(a) $= 5$; (b) $= -2$; (c) $= 3$; (d) $= 7$

2 **[Foundational]** • **Target: Evaluating logarithms**

Use the change-of-base formula to evaluate $\log_5 18$, to three decimal places.

$\log_5 18 = \frac{\ln 18}{\ln 5} \approx \frac{2.8904}{1.6094} \approx 1.796$.

3 **[Foundational]** • **Target: Evaluating logarithms**

State the domain of $f(x) = \log(x - 4)$ and the inverse function of $g(x) = 2^x$.

Domain of f: $x > 4$. Inverse of g: $g^{-1}(x) = \log_2 x$.

4 **[Foundational]** • **Target: Properties of logarithms**

Expand fully using properties of logarithms: $\ln\left(\frac{x^3\sqrt{y}}{z^2}\right)$.

$3\ln x + \frac{1}{2}\ln y - 2\ln z$.

5 **[Foundational]** • **Target: Properties of logarithms**

Write as a single logarithm: $2\log x + 3\log y - \log z$.

$\log\left(\frac{x^2y^3}{z}\right)$.

6 **[Foundational]** • **Target: Properties of logarithms**

Given $\log_b 2 = 0.431$ and $\log_b 3 = 0.683$, evaluate:

a $\log_b 6 = 1.114$ **b** $\log_b 8 = 1.293$ **c** $\log_b 1.5 = 0.252$

(a) $= 1.114$; (b) $= 1.293$; (c) $= 0.252$

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7 [Foundational] • Target: Graphs of logarithmic functions

For a curriculum assessment problem, suppose for $y = \log_2(x+4) - 1$, state:

a the domain $x > -4$ **b** the vertical asymptote $x = -4$ **c** the x-intercept $0 = \log_2(x+4) - 1 \Rightarrow x = -2$

(a) $x > -4$; (b) $x = -4$; (c) $0 = \log_2(x+4) - 1 \Rightarrow x = -2$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Graphs of logarithmic functions

Describe how the graph of $g(x) = -\ln(x-2) + 3$ is obtained from $y = \ln x$.

Shift right 2 units, reflect across the x-axis, then shift up 3 units.

9 [Intermediate] • Target: Logarithmic scales and applications

The Richter scale measures earthquake magnitude by $M = \log\left(\frac{I}{I_0}\right)$, where I is intensity and I_0 is a reference intensity.

a An earthquake has $M = 6$. Express I in terms of I_0. $I = I_0 \cdot 10^6$.

b How many times more intense is a magnitude-7 earthquake than a magnitude-5 earthquake? Difference in magnitude is 2, so intensity ratio is $10^2 = 100$ times.

(a) $I = I_0 \cdot 10^6$.; (b) Difference in magnitude is 2, so intensity ratio is $10^2 = 100$ times.

10 [Intermediate] • Target: Logarithmic scales and applications

Sound intensity level in decibels is $\beta = 10\log\left(\frac{I}{I_0}\right)$. If a sound's intensity doubles, by how much does β increase, to two decimal places?

$\Delta\beta = 10\log(2) \approx 10(0.3010) \approx 3.01$ dB.

11 [Intermediate] • Target: More practice with logarithm properties

Expand fully: $\log_2\left(\frac{8x^2}{\sqrt[3]{y}}\right)$.

$\log_2 8 + 2\log_2 x - \frac{1}{3}\log_2 y = 3 + 2\log_2 x - \frac{1}{3}\log_2 y$.

12 [Intermediate] • Target: More practice with logarithm properties

Write as a single logarithm and simplify: $\frac{1}{2}\ln 16 + \ln 3 - \ln 2$.

$\ln(16^{1/2}) + \ln 3 - \ln 2 = \ln 4 + \ln 3 - \ln 2 = \ln\left(\frac{4\cdot 3}{2}\right) = \ln 6$.

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13 [Intermediate] • Target: Logarithms as inverse functions

The graphs of $y = 2^x$ and $y = \log_2 x$ are shown together with the line $y = x$. Explain why the two curves are reflections of each other across $y = x$.

Since $y = \log_2 x$ is defined as the inverse of $y = 2^x$, swapping the roles of x and y (which is exactly what reflecting across the line $y = x$ does geometrically) transforms one graph into the other.

14 [Intermediate] • Target: Solving simple logarithmic equations

Solve $2\log x = \log(x + 6) + \log 1$ for x (noting $\log 1 = 0$), and check for extraneous solutions.

$2\log x = \log(x+6) \Rightarrow \log(x^2) = \log(x+6) \Rightarrow x^2 = x + 6 \Rightarrow x^2 - x - 6 = 0 \Rightarrow (x-3)(x+2) = 0$. $x = 3$ or $x = -2$. Since $\log x$ requires $x > 0$, $x = -2$ is extraneous. Solution: $x = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: The pH scale (a logarithmic application) • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

pH is defined as $pH = -\log[H^+]$, where $[H^+]$ is hydrogen ion concentration (mol/L). Find the pH of a solution with $[H^+] = 3.2 \times 10^{-5}$, to two decimal places.

$pH = -\log(3.2 \times 10^{-5}) \approx -(-4.4949) \approx 4.49$.

16 [Challenging] • Target: The pH scale (a logarithmic application) • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Two solutions have pH values of 3 and 6. Find the ratio of their hydrogen-ion concentrations $[H^+]$, and state which is more acidic.

$[H^+]_1 = 10^{-3}$, $[H^+]_2 = 10^{-6}$. Ratio $= \frac{10^{-3}}{10^{-6}} = 10^3 = 1000$. The pH-3 solution has 1000 times the hydrogen-ion concentration and is far more acidic (lower pH means more acidic).

17 [Challenging] • Target: Common mistakes with logarithm properties • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Explain why $\log(x + y) \neq \log x + \log y$ in general, using a specific numerical counterexample.

Using $x = y = 10$: $\log(10 + 10) = \log 20 \approx 1.301$, but $\log 10 + \log 10 = 1 + 1 = 2$. Since $1.301 \neq 2$, the two expressions are not equal — the correct property is $\log x + \log y = \log(xy)$, not $\log(x + y)$.

18 [Challenging] • Target: Common mistakes with logarithm properties • Cross-Lesson: 3.03 Sum and Difference Formulas

Simplify $\log_2(4x) - \log_2(2x)$ using the quotient property, and verify with a specific value of x.

$\log_2(4x) - \log_2(2x) = \log_2\left(\frac{4x}{2x}\right) = \log_2(2) = 1$. Check with $x = 5$:
$\log_2(20) - \log_2(10) \approx 4.32 - 3.32 = 1$ ✓.

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19 [Challenging] • Target: Evaluating logarithms • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

[Functions & Their Graphs · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

20 [Challenging] • Target: Properties of logarithms • Cross-Lesson: 4.01 Circles and Parabolas

[Unit 2 · Exponential and Logarithmic Functions · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

Substitute $t = 10$: $g(10) = 6(10) + 6 = \mathbf{66}$. Setting $6t + 6 = 72$ gives $6t = 66 \Rightarrow t = \mathbf{11}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Evaluating logarithms

Consider the functions $f(x) = \frac{1}{x-2}$ and $g(x) = 3x + k$. If $f(g(x)) = \frac{1}{2x-1}$, find the value of k.

a. 3

b. 2

c. **1** ✓

d. -1

We are given the functions $f(x) = \frac{1}{x-2}$ and $g(x) = 3x + k$. We are also given that $f(g(x)) = \frac{1}{2x-1}$. To find $f(g(x))$, we substitute $g(x)$ into $f(x)$: $f(g(x)) = f(3x + k) = \frac{1}{(3x+k)-2} = \frac{1}{3x+k-2}$. Now, we equate this to the given expression for $f(g(x))$: $\frac{1}{3x+k-2} = \frac{1}{2x-1}$. For these two fractions to be equal, their denominators must be equal: $3x + k - 2 = 2x - 1$. To solve for k, we need to rearrange the equation. It seems there might be a mistake in the.

22 [Foundational] • Target: Properties of logarithms

Let $f(x) = \log_2(ax + b)$ and $g(x) = 3^x - 1$. Given that $f(g(x)) = x$ and the point $(2, 1)$ lies on the graph of $y = g(x)$, find the value of $a + b$.

a. **4** ✓

b. 3

c. 2

d. 1

We are given the function $f(x) = \log_2(ax + b)$. We are also given a function $g(x)$ and a point $(2, 1)$ that lies on its graph, and the condition $f(g(x)) = x$. The condition $f(g(x)) = x$ implies that f is the inverse function of g, i.e., $f(x) = g^{-1}(x)$. Let's first use the information about the point $(2, 1)$ and the structure of $g(x)$. If $g(x) = 2^x + k$ (assuming a general form for exponential functions with base 2 to match the logarithm base), and the point $(2, 1)$ lies on it, then: $g(2) = 1$.

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23 [Foundational] • Target: Graphs of logarithmic functions

Given the function $f(x) = 2x + 3$. What is the value of $f^{-1}(7)$?

a. 1

b. **2** ✓

c. 3

d. 4

Let $y = f(x)$. Then $y = 2x + 3$. To find the inverse function $f^{-1}(y)$, we swap x and y and solve for y. So, $x = 2y + 3$. Subtract 3 from both sides: $x - 3 = 2y$. Divide by 2: $y = \frac{x-3}{2}$. Thus, $f^{-1}(x) = \frac{x-3}{2}$. We need to find $f^{-1}(7)$. Substitute $x = 7$ into the inverse function: $f^{-1}(7) = \frac{7-3}{2} = \frac{4}{2} = 2$.

24 [Foundational] • Target: Logarithmic scales and applications

Consider the functions $f(x) = 3x + 2$ and $g(x) = x - 5$. What is the value of $f(g(3))$?

a. **1** ✓

b. -4

c. 13

d. -10

To find $f(g(3))$, we first evaluate the inner function $g(3)$. $g(x) = x - 5$ $g(3) = 3 - 5 = -2$ Now we substitute the result of $g(3)$ into the outer function $f(x)$. $f(x) = 3x + 2$ $f(g(3)) = f(-2) = 3(-2) + 2$ $f(-2) = -6 + 2$ $f(-2) = -4$ Therefore, $f(g(3)) = -4$.

25 [Foundational] • Target: More practice with logarithm properties

Which is the inverse of $f(x) = \log_2 x$?

a. $g(x) = \log_2 x$

b. $g(x) = 2x$

c. $g(x) = x^2$

d. $g(x) = 2^x$ ✓

Applying Exponential & Logarithmic Functions principles, Exponential base 2. Therefore, the verified correct option is $g(x) = 2^x$.

26 [Foundational] • Target: Logarithms as inverse functions

If $f(x) = x^2 + 1$ and $g(x) = -x - 1$, find the sum of all solutions to $(f \circ g)(x) = 5$.

a. **-2** ✓

b. 4

c. -6

d. -10

$(-x-1)^2 + 1 = 5 \Rightarrow (-x-1)^2 = 4 \Rightarrow -x - 1 = \pm 2 \Rightarrow x = -3$ or 1; sum $= -2$.

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27 [Foundational] • Target: Solving simple logarithmic equations

Find the number of real solutions to $f(x) = x$ where $f(x) = x^3 - 3x$.

a. 1

b. 0

c. 3 ✓

d. 2

$x^3 - 3x = x \Rightarrow x^3 - 4x = 0 \Rightarrow x(x-2)(x+2) = 0 \Rightarrow x = 0, 2, -2$. Three real solutions.

28 [Foundational] • Target: The pH scale (a logarithmic application)

Find the number of surjections from a 4-element set to a 2-element set.

a. 14 ✓

b. 8

c. 16

d. 12

Total functions $2^4 = 16$, subtract non-surjective (1 onto each singleton: $1 + 1 = 2$). So $16 - 2 = 14$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Common mistakes with logarithm properties

Find all functions f with $f(x)^2 = x^2$ for all x.

a. $f(x) = x$, $f(x) = -x$, **or** $f(x) = |x|$ **or** $f(x) = -|x|$ ✓

b. $f(x) = x$ only

c. $f(x) = 0$

d. $f(x) = \pm 1$

$f(x)^2 = x^2 \Rightarrow f(x) = \pm|x|$. For each x, $f(x)$ can independently be $+|x|$ or $-|x|$.

30 [Intermediate] • Target: Evaluating logarithms

Find all polynomials $p(x)$ of degree 2 satisfying $p(x+1) - p(x) = 2x + 1$ and $p(0) = 0$.

a. $p(x) = 2x$

b. $p(x) = x^2 - x$

c. $p(x) = x^2 + x$

d. $p(x) = x^2$ ✓

Let $p(x) = ax^2 + bx + c$. $p(x+1) - p(x) = 2ax + a + b = 2x + 1 \Rightarrow a = 1, b = 0$. $p(0) = c = 0$. So $p(x) = x^2$.

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31 [Intermediate] • Target: Properties of logarithms

Solve: $3^x = 2^{x+1}$. Use logarithms.

a. $x = \frac{\ln 2}{\ln 3}$

b. $x = 1$

c. $x = \frac{\ln 3}{\ln 2}$

d. $x = \frac{\ln 2}{\ln 3 - \ln 2}$ ✓

$x\ln 3 = (x+1)\ln 2 \Rightarrow x(\ln 3 - \ln 2) = \ln 2 \Rightarrow x = \frac{\ln 2}{\ln 3 - \ln 2}$.

32 [Intermediate] • Target: Graphs of logarithmic functions

What value must be excluded from the domain of $f(x) = \frac{1}{3x+3}$?

a. 3

b. **-1** ✓

c. -4

d. 2

Denominator $= 0 \Rightarrow 3x + 3 = 0 \Rightarrow x = -1$, which must be excluded.

33 [Intermediate] • Target: Logarithmic scales and applications

State and apply the intermediate value theorem: if $f(1) = -3$ and $f(4) = 5$, continuous on $[1, 4]$, then:

a. No conclusion possible

b. **There exists** $c \in (1, 4)$ **with** $f(c) = 0$ ✓

c. $f(2) = 1$

d. f is constant

IVT: continuous function on $[a, b]$ attains every value between $f(a)$ and $f(b)$.

34 [Intermediate] • Target: More practice with logarithm properties

If $f(x) = -2x + 1$ and g is the inverse of f, find $g(f(-1))$.

a. 4

b. **-1** ✓

c. 2

d. -10

By definition of inverse functions, $g(f(x)) = x$ for all x, so $g(f(-1)) = -1$.

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35 [Intermediate] • Target: Logarithms as inverse functions

Find a value of c making f continuous at $x = 2$, where $f(x) = \{cx + 1, x < 2; x^2 - 1, x \geq 2$.

a. $c = 1$ ✓

b. $c = 3$

c. $c = 0$

d. $c = 2$

$\lim_{x \to 2^-} (cx + 1) = 2c + 1$. $f(2) = 3$. Set $2c + 1 = 3 \Rightarrow c = 1$.

36 [Intermediate] • Target: Solving simple logarithmic equations

The Banach fixed-point theorem guarantees a unique fixed point when f is:

a. Any continuous function

b. Lipschitz with $L = 1$

c. A bijection

d. **A contraction mapping on a complete metric space** ✓

Banach FPT: contraction ($L < 1$) on complete space has unique fixed point.

37 [Intermediate] • Target: The pH scale (a logarithmic application)

Solve: $2^{x+1} = 5$. Express x using logarithms.

a. $x = \frac{\ln 5}{\ln 2} - 1$

b. $x = \frac{\ln 5 - 1}{\ln 2}$

c. $x = \log_2 5 - 1$

d. **Both A and C** ✓

$x + 1 = \log_2 5 \Rightarrow x = \log_2 5 - 1 = \ln 5 / \ln 2 - 1$. A, C are equivalent.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Common mistakes with logarithm properties • Cross-Lesson: 3.03 Sum and Difference Formulas

Find all rational functions $f(x) = \frac{ax+b}{cx+d}$ that are their own inverse.

a. $b = c$

b. $f(f(x)) = x$ impossible for rational

c. $a = d$

d. $a = -d$ **(i.e.** $a + d = 0$**)** ✓

$f(f(x)) = x \Rightarrow$ involutory Möbius transform: condition $a + d = 0$.

39 [Challenging] • Target: Evaluating logarithms • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

What is the smallest integer in the domain of $f(x) = \sqrt{2x+2}$?

a. -8

b. -9

c. **-1** ✓

d. -10

Need $2x + 2 \geq 0 \Rightarrow x \geq -1$; smallest integer $= -1$.

40 [Challenging] • Target: Properties of logarithms • Cross-Lesson: 4.01 Circles and Parabolas

If $f(x) = \sqrt{9 - x^2}$, what is the shape of its graph?

a. Parabola

b. **Upper semicircle of radius 3** ✓

c. Full circle

d. Hyperbola

$y = \sqrt{9 - x^2} \Rightarrow x^2 + y^2 = 9$, $y \geq 0$: upper semicircle.

41 [Challenging] • Target: Graphs of logarithmic functions • Cross-Lesson: 4.02 Ellipses

Find the interval where $f(x) = x^2 - 4x + 3$ is positive.

a. $x > 0$

b. $1 < x < 3$

c. $x < 0$

d. $x3$ ✓

Roots at $x = 1, 3$. Upward parabola positive outside roots.

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42 [Challenging] • Target: Logarithmic scales and applications • Cross-Lesson: 4.03 Hyperbolas

The Weierstrass function is continuous everywhere and:

a. Differentiable everywhere

b. Differentiable at integers only

c. **Differentiable nowhere ✓**

d. Piecewise linear

Classic example: continuous but nowhere differentiable.

43 [Challenging] • Target: More practice with logarithm properties • Cross-Lesson: 5.01 Vectors: Components and Operations

Which describes a vertical stretch of $y = f(x)$ by factor 4?

a. $y = f(4x)$

b. $y = f(x/4)$

c. $y = f(x) + 4$

d. $y = 4f(x)$ ✓

Vertical stretch: multiply output by constant.

44 [Challenging] • Target: Logarithms as inverse functions • Cross-Lesson: 5.02 The Dot Product

Which function is the inverse of $f(x) = e^x$?

a. $g(x) = x^e$

b. $g(x) = \ln x$ ✓

c. $g(x) = 10^x$

d. $g(x) = \log x$

Natural log is inverse of natural exponential.

45 [Challenging] • Target: Solving simple logarithmic equations • Cross-Lesson: 5.03 Parametric Equations

A function has zeros at $x = -1, 0, 2$ and positive leading coefficient. Sketch shows which end behavior?

a. Left: $f \to +\infty$; Right: $f \to -\infty$

b. **Left:** $f \to -\infty$**; Right:** $f \to +\infty$ ✓

c. Both ends to $+\infty$

d. Both ends to $-\infty$

Odd degree (3), positive leading coefficient.

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